docs/encrypted-data.md)](https://typographics-art.org/docs/encrypted-data.md) and [do
e [docs/encrypted-data.md](https://typographics-art.org/docs/encrypted-data.md) and [do
one receivers everything; a strict 2-of-3 mode exists), with Argon2id for the extra pass
a different cipher is not the answer). A plain-language page for
md) (German: [docs/passwords-and-keys.de.md](https://typographics-art.org/docs/passw
today's cryptography is known", not "nobody can ever get at it": a lost key lo
ypted media and the adapter.** A picture or a video can be sealed so it only opens with the
without the plaintext ever touching the disk and without anything being uploaded. Large m
me, `size`. See [docs/encrypted-media.md](https://typographics-art.org/docs
view-internal-2026-10.md); a second review by a different model, done blind to the first, is
ler text (for example Lorem ipsum for a finer picture) is allowed only if it is declared. 2. **T
ll definition: [docs/technique.md](https://typographics-art.org/docs/technique.md). ## I
hoto colour] C --> O[HTML + SVG + PDF
text stays text] ``` 1. **Measure.** For each char
size (and width, if enabled) whose ink best matches the area it covers. The leftover error is
erage to the photo colour. Details and formulas: [docs/algorithm.md](https://typographics-
Poster. Export HTML, SVG, PNG, or print to PDF. Your text and photo are processed in
DFs embed their fonts. The page [**/wo
ny **dependencies. ES modules for the
e color: 'photo', // photo | gray | mono p
racters toSvg(sheet); // SVG, text as c
2> (the CLI uses the global `WebSock
hoto.jpg --text poem.txt --fill 12000 -o
t], paid with a DigiByte transaction (no a
a Gioconda**. • Walter Pater · Leonarde
ps://typographics-art.org/gallery/phot
(https://github.com/typ0graphic5/tg-ar
llery`. ## Where this is useful - **Arth
ics-art.org/docs/technique.md), rule 2
tay optical size, slant), - typeface, a
ny characters are filler, and the filler a
ub.com/typ0graphic5/tg-art/blob/main/
main/CONTRIBUTING.md); please follow
m) # only to deploy the website: t
--] | `src/` | the library | | `cli/` | comman
ements and rules | | `.claude/` | gener
, the project's own texts and motifs an
ng a picture from a given text by setting
t follows three rules. ### 1. Fidelity
gap at a space (or indent a line) so that
en (a URL, a base64 string, the dot-join
aracters as text: HTML, SVG, or a PDF
o nearly one per glyph; concatenated th
he HTML `role="figure"` label). Exporte
, optical size, slant), - typeface, a
o the eye at viewing size. Very fine wor
touch the rules above: the work still co
gorithm.** ## How the setting is wor
h the browser lays the glyph out with, m
pset()` loads the font's stylesheet firs
an unsharp mask (**emphasise edges**
× (1 − grey) × lineRatio / 0.92 ``` For eve
ext to full-size ones read like stray cap
dge tool sets the tone of the side its set again there. In word mode, one continuous
heet (3 000 characters make about 42 li
scanned in steps of 0.3 font sizes; stretches w
mages too. - **Fidelity**: `isFaithful(sheet)` e
at its works stay readable for machines. T
. The filler contains words like **official**, wh
all `<text>` content, whitespace normalised).
ity | | image only | `tesseract` on a PNG render
--|---|---|---|---| | high | clear | exact | exact
ge model (first sample) Classic OCR knows only
the word order cannot be known in advance. Me
m | 65 | 100 % | | free | 94 | 100 % | This is a small
Fe.** Chrome set **ffi** in *official* as one ligature gly
, and `pdftotext`'s layout mode moved the very first w
Moby-Dick Master The determinator in `demos/moby-dick/
draw -F txt`) returns every visible character. The cause
e the PDF text layer. The SVG master keeps them; the PD
tors read a one-pager completely Measured on a 1 200 × 1
g order | | pdfminer.six | 10 of 10 | | | pypdf | 10 of 10 | reads a
ors above that lose most of their text (10–30 % read). Its defau
ze), this holds even for the EU AI Act at 0.5 errors per
e as not exact. The agent library avoids arrows for that reason.
stored order; with legibility *high* it is exact for layout-reconstruc
than 50 000 small characters. Tools that build the visual layout c
reproducible numbers are very welcome; open an issue or pull requ
ge change the picture. All numbe come from `scripts/size-study.m
ght × width × type size** (measu as 5 sizes gave 180 f
nes × 7.2 B × runs × 0.9 B × letter 4 KB after slimming = 328 B × outline
ed without visible change | Step ng | Why it is safe | |---|---|---| Rou
one empty `/NonStruct` element r run under one `/Document` — the
set to (title or motif card), both b 1.5 mm, in brightness and in colou
change to full | |---|---|---| | | s, widths and sizes, colour step 6, to
, one width | 1.93 MB | 40 | 0.942 / % | | 2 weights, one width | 1.52 MB
| | colour step 96, one width | 2.6 0.931 / 0.918 | 1.39 % | | 2 weights, 1 s
s brightness / colour | change to f |---|---| all weights, widths and sizes, c
| 4 weights, one width | 0.58 MB | | 53 | 0.60 % | | 2 weights, one width | 0.
5 | 0.30 % | | colour step 96, one wi 7 | 0.998 / 0.995 | 0.30 % | | 2 weights,
tal streak; with one size the photo motif follows the picture more closely (0.965 inst
- **Width variants are not worth it.** Dropping condensed and expanded changes no
e.mjs`); `--weights 2` makes them smaller still. In the app, the font-cuts slider limits t
=study.mjs --evaluate # measures the PDFs of the last run again ``` ## Hidden message
e key. ## How it works It is a book cipher over the work's public text. 1. **The public t
th MuPDF). 2. **Hiding:** for every character of the secret, all its occurrences in the publ
and refused Instead of producing garbage. 4. **Brute force** the characters at the listed pos
t on top of a short header: measured on a 47 000-character text, 26 characters of secret ga
ecret, so it is better not to put it on the command line, where it shows up in the process list an
org/app), the panel **Hidden message** does the same for the current work. In code: `hideSecr
secret.** Whoever has the key and the work can read the message. Share the key over anothe
the message must stay secret even from someone who may get the key: then encrypt the messa
sent from short ones. - The key files exactly one text. File-setting the same text into a new picture keep
its source and the key made from t `docs/ai-readability.md`). ## Hone
md --> ## Passwords, keys and seed w it works here, and what we do not
od. Everything is open source and s or is marked as an estimate o
s/key-management.md) and [cipher /docs/cipher-review.md). A German
e did not invent a cipher, and we look worth the risk of adding code. - **A
efuse to publish a record e key we cannot recover is.** The ga
and nobody can help you. - **It is not dard; the code around it (formats, bin
n sizes vary a lot. ## Not measured yet - -language models
--> ## File size, measured # File size: makes a PDF big, and
e-study.json`) on two one-pagers, pr with Chrome 154 and r
rge sheets printed in several strips carry their text once per
letters × 7 KB *outlines* are distinct weight × width × size × c
e units (`cli/pdf-slim.js`) | −22 % (Swiss constitution) | at most
ext and order are unchanged | | Repack with qpdf | −0 to −8 % |
e to full* is the plain difference to the version with every varian
82 | 0.940 / 0.924 | 0.00 % | | as published so far (colour step 24) |
28 % | | 1 weight, one width (a single font) | 1.34 MB | 10 | 0.943 /
MB | 4 | 0.946 / 0.939 | 1.82 % | | 1 weight, 2 sizes, one width | 1.07
PDF | 0.73 MB | 22 | 0.995 / 0.995 | 0.00 % | | as published so far (c
0.977 | 0.77 % | | 1 weight, one width (a single font) | 0.68 MB |
0.54 MB | 4 | 0.994 / 0.994 | 0.42 % | | 1 weight, 2 s
hird of the size of the full version. On the title
and saves 12 %. - **Coarser colours har
hows the expected PDF size. Works ma
--> ## Hidden messages # Hidden ma
he way a machine reads the work: wh
cted and one is chosen at random (W
are the secret. ## The key `tg1.` foll
06 characters; a 30-character secret g
ll history: `reveal` also takes it from a
, secret)` and `revealSecret(public
and keep it like a password. - **Rando
nd hide the ciphertext's characters, o
valid (the text is the same); changing
does not touch the [rules of the tech
e This page is for people who want to
clusion, and the last section show
on of this page: [passwords-and-key
word is the weak point of every lock i
oes its work in your browser and doe
ge, key handling) was written for this pr
or the full text layout-reconstructing ex
moby-dick/pred.mjs`). Three findin
text is in the file; the proof extracts
s, sizes and light colours, Tesseract
haracters counted, exact text equal
4 %) | | Poppler `pdftotext` (default 7
| Poppler `pdftotext -r 600` | 10 of 10
type. Text is not font, and MuPDF th
ers in an ordinary font shows the sa
out 99 % of the words. - **PDF extra
on and whole pages instead of cropping. - Other OCR e
s can be seen A Typographics Art PDF carries the whole tex
back exactly in every case. ## Where the bytes are Chrom
ge sheets printed in several strips carry their text once per
acter, *runs* the style changes (`pdfStats` and `estimateP
/000 of the type size; at 600 dpi no pixel changed by more t
e constitution −5 %, Basic Law −80 %) | losseless (object str
ll rows include the slimming steps above; from "one width"
s 3 MB | 193 | 0.939 / 0.933 | 0.25 % | | without accessibility ta
45 | 3.32 % | | 2 sizes, one width | 1.92 MB | 40 | 0.949 / 0.935
| 4 | 0.951 / 0.951 | 3.40 % | | 1 weight, 1 size, one width: only
ur step 24) | 0.73 MB | 22 | 0.995 / 0.995 | 0.00 % | | without acc
72 / 0.982 | 2.20 % | | 2 sizes, one width | 0.55 MB | 12 | 0.995 /
0.979 | 2.72 % | | 1 weight, 1 size, one width: only
carries the brightness.** Two weigh
of 24 save 4 %; step 48 looks the sam
the texture of mixed sizes is a choic
ssage that only the holder of a key
soft hyphens, direction marks) remo
st of chosen positions. 3. **Fingerpr
n the positions as variable-length inte
g-secret.mjs hide work.pdf --secret "M
ey-stdin`) or from the environment variable `TG_S
cts, and value 1 does not. • **The work reveals not
ey every time, and repeated letters of the secret p
hat does not occur in the public text cannot be hid
ne PDF quirk: after a character drawn from a fall
md): the work contains its declared text and noth
Art protects data, without reading the cryptogra
cuments behind it are [encrypted-data.md
-and-keys.de.md). ## 1. In one minute - **The
public or permanent, the real protection is a
site reads nothing from third parties, and t
ryptographer has reviewed it. If losing o
words, or lxe) | The one backup fr